SRI GANESH COLLEGE OF ENGINEERING & TECHNOLOGY

and

PSV COLLEGE OF ARTS & SCIENCE

Gladly Presents,

Free Summer Course on AI & Ethical Hacking

DAY 3

ARTIFICIAL INTELLIGENCE

WHAT IS AI?

- Artificial Intelligence (AI) is a multidisciplinary field of computer science focused on creating systems that can perform tasks that would typically require human intelligence.
- These tasks include reasoning, learning, problem-solving, perception, language understanding, and decision-making.
- AI aims to develop machines that can mimic cognitive functions such as learning from experience, adapting to new situations, and understanding natural language.
- **Examples:** Google Assistant, Self-driving cars, ChatGPT, Netflix recommendations.

TYPES OF AI

- **Narrow AI:** AI systems that are designed for a specific task.

  Example: Siri, Alexa

- **General AI:** AI with human-level cognitive abilities.

  Not yet achieved

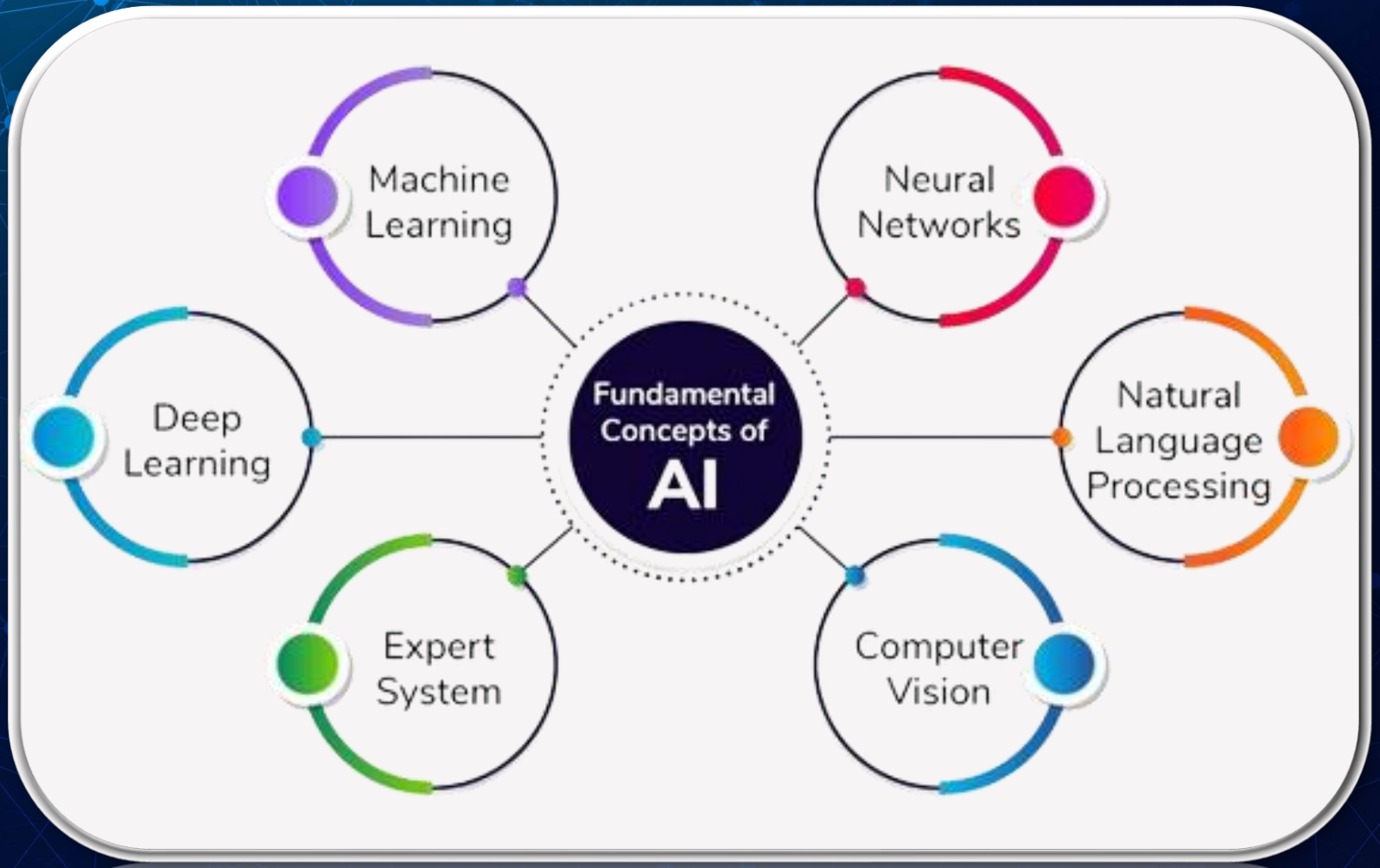
- **Super AI:** AI that surpasses human intelligence.

  Theoretical at this stage

HOW AI WORKS?

- **Step 1: Data Collection** – Collecting data from various sources.
- **Step 2: Data Preprocessing** – Cleaning and organizing the data.
- **Step 3: Model Training** – Using algorithms to learn patterns.
- **Step 4: Prediction** – Making decisions based on new input.
- **Step 5: Improvement** – Retraining with more data for better results.

COMPONENTS OF AI



- **Machine Learning (ML):** Teaching computers to learn from data.
- **Natural Language Processing (NLP):** Understanding human language.
- **Computer Vision:** Understanding images and videos.
- **Robotics:** Physical machines controlled by AI.
- **Expert Systems:** Rule-based decision-making systems.

INTRODUCTION TO MACHINE LEARNING

- A subset of AI that enables machines to learn from data and improve without being explicitly programmed.
- ML algorithms find patterns in data and make decisions or predictions.

TYPES OF MACHINE LEARNING

- **Supervised Learning:** Learns from labeled data

  Example:

  - Spam Email Detection
  - Predicting house prices based on features

- **Unsupervised Learning:** Works with unlabeled data to find hidden patterns or groupings

  Example:

  - Customer Segmentation
  - Market Basket Analysis

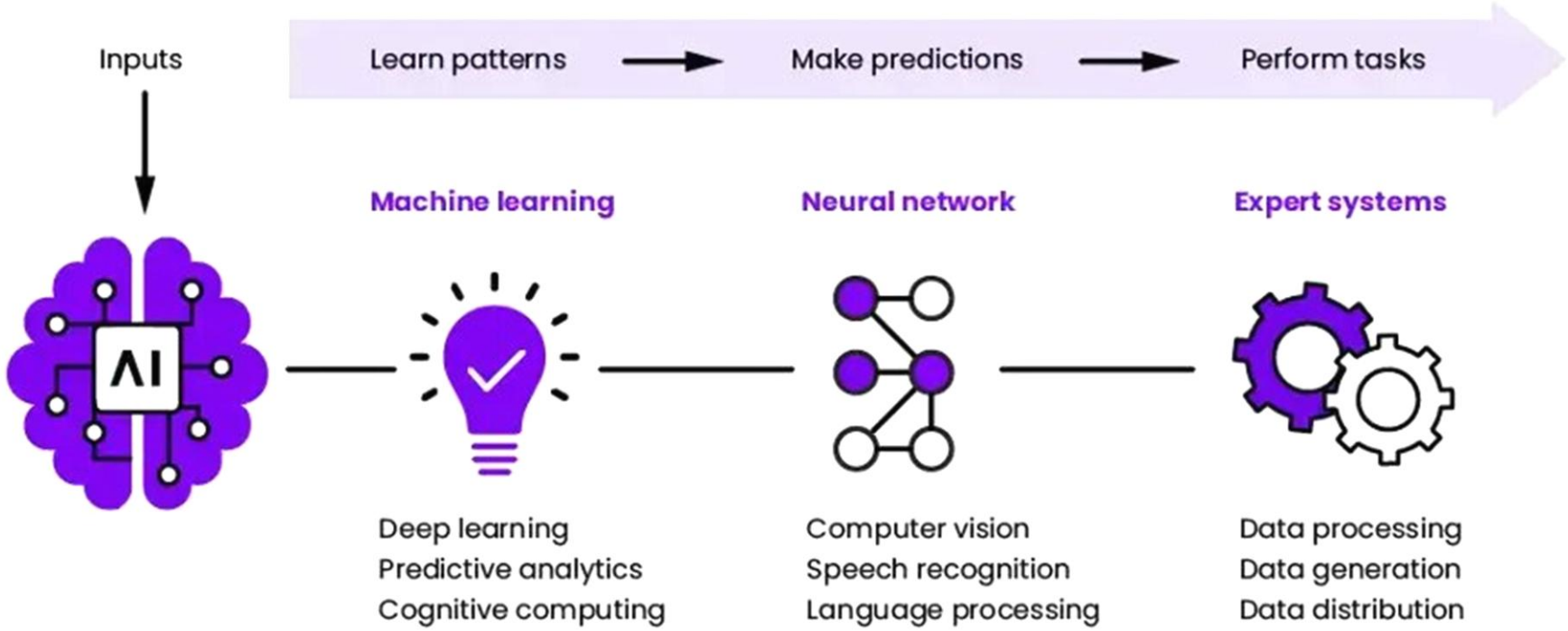
- **Reinforcement Learning: Learns by trial and error through rewards and penalties**

Example:

- **Game-playing AI (like AlphaGo)**
- **Self-driving cars learning to avoid obstacles**

PROS & CONS OF AI

PROS	CONS
➢ Helps with Tasks ➢ Automation ➢ Assistance	➢ Job Displacement ➢ Bias and Fairness ➢ Privacy Concerns ➢ Dependence and Control


Inputs
Learn patterns
Make predictions
Perform tasks
AI
Machine learning
Neural network
Expert systems
Deep learning
Predictive analytics
Cognitive computing
Computer vision
Speech recognition
Language processing
Data processing
Data generation
Data distribution

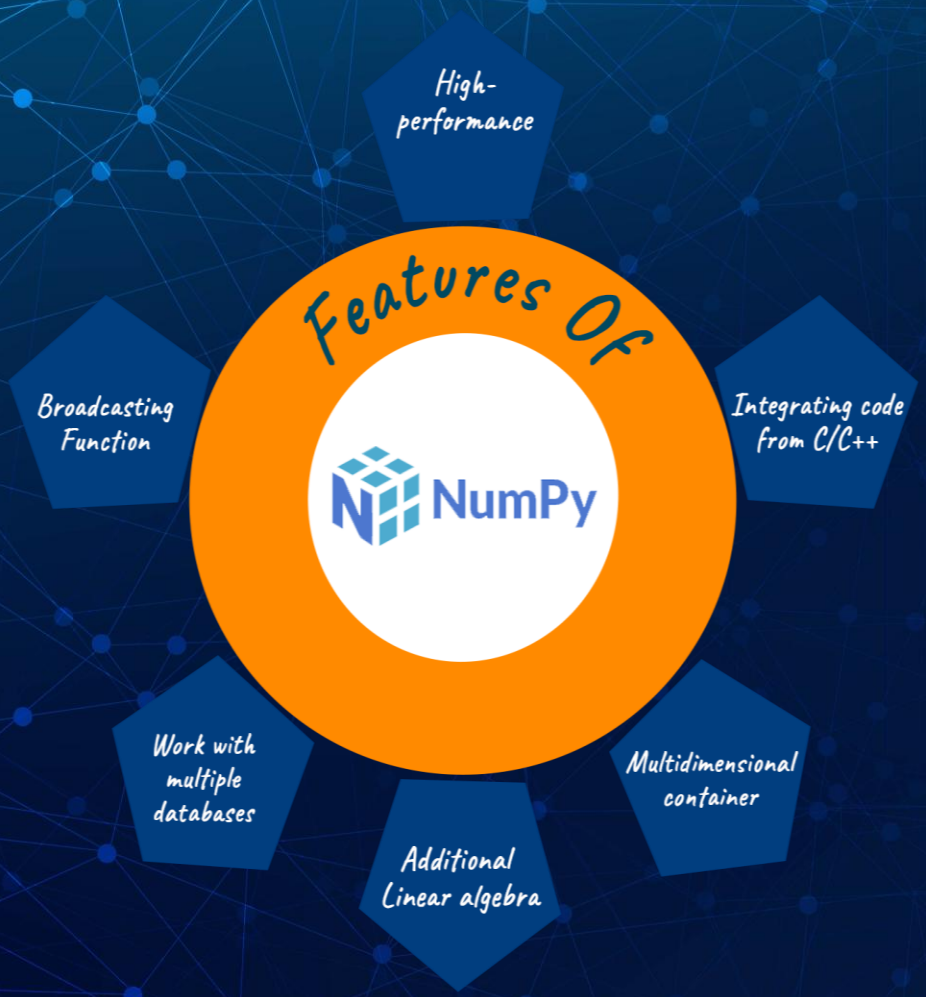
LIBRARIES IN PYTHON

- A library is a collection of pre-written code that helps you do specific tasks without writing everything from scratch.

Most used Libraries:

- NumPy
- Pandas
- Matplotlib
- TextBlob

1. NumPy



- NumPy - (Numerical Python)
- Used for numerical operations
- Supports multi-dimensional arrays (ndarrays)
- Fast and memory-efficient

Math with arrays and numbers

2. Pandas

- Used for data manipulation and analysis
- Works with tables (Data Frames)
- Easy to load, filter, and modify data

Working with tables and data

3. MathPlotlib

- Used for data visualization
- Create graphs and plots easily
- Works well with Pandas & NumPy
- Drawing charts and graphs

matplotlib

Drawing charts and graphs

4. TextBlob

- TextBlob is a Python library for processing textual data.
- It makes it super easy to perform Natural Language Processing (NLP) tasks like:
  - Sentiment Analysis
  - Language Detection
  - Translation
  - Spell Correction
  - Noun Phrase Extraction

Analyzing text (sentiment, language)

THANK YOU